

# TIC-TAC-TOE GAME

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# Using REINFORCEMENT LEARNING

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# Project Overview



## Tic-Tac-Toe AI Game Using Q-Learning

Develop a Tic-Tac-Toe game where a human plays against a Q-learning AI. The game manages valid moves and board states. Players compete in a best-of-three format.

The AI learns through self-play, improving its strategy. The interface uses Pygame for simplicity and ease of use. This project demonstrates reinforcement learning in an engaging, educational way.



# X CORE FEATURES

## Game State & Match Flow:

Validates moves, resets board, tracks scores, and controls best-of-three matches.

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## Simple AI Move Selection:

AI tries to block or win, prioritizes center and corners, and switches turns fairly.

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## Minimal Web UI Experience:

Interactive HTML/JS frontend updates board and scores instantly, with easy round and match resets.





# Win Rate Percentages

Q-learning AI

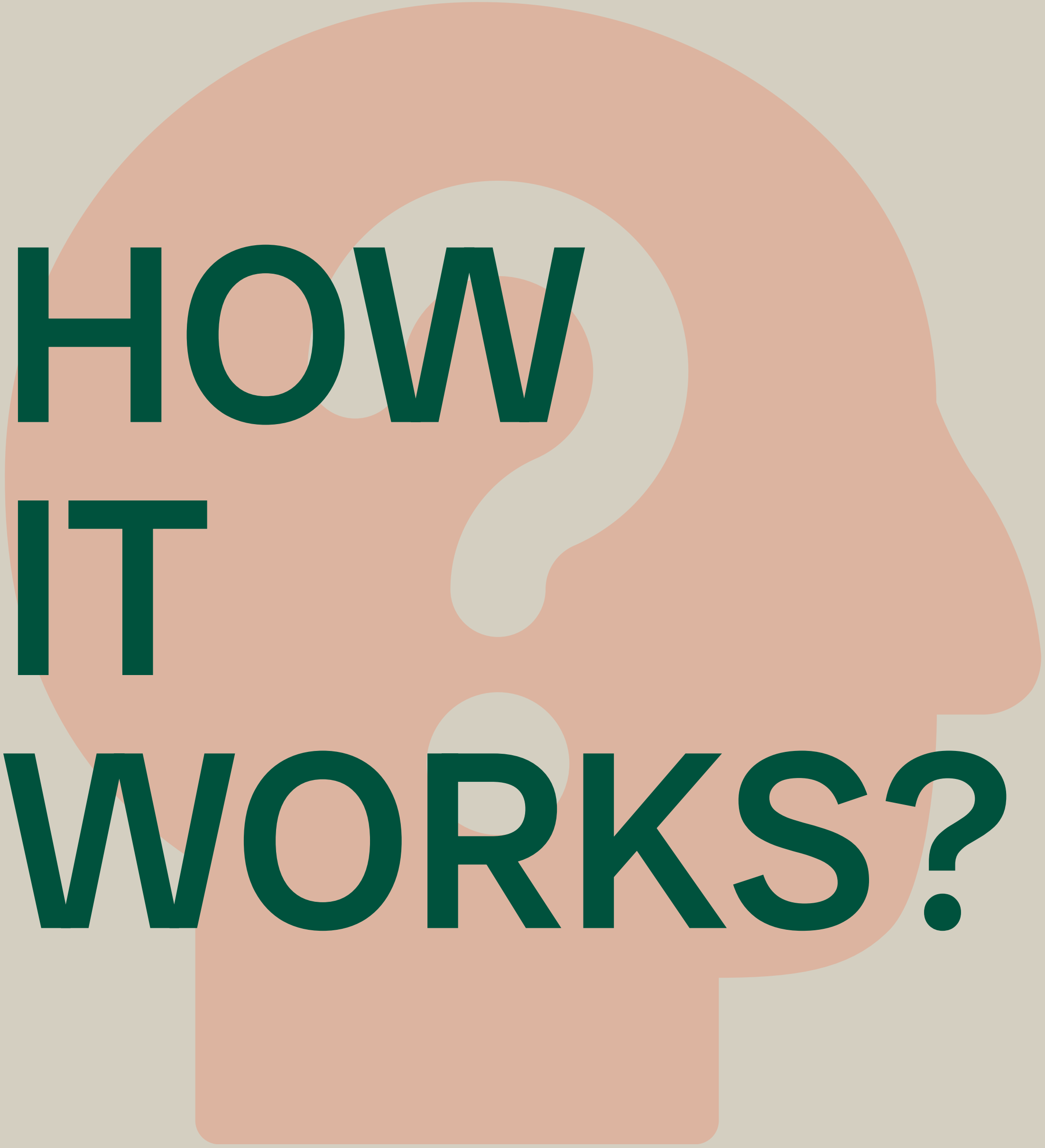
59.5%

Minimax AI vs  
Random

94.5%

Policy Gradient AI

49.5%



# HOW IT WORKS?

**01.**

## **Training Phase:**

The AI learns by playing against itself repeatedly, improving its strategy through reinforcement learning.

**02.**

## **Gameplay Phase:**

During live matches, the AI quickly selects its best moves while the human plays by clicking on the board.

**03.**

## **Match Tracking:**

The system tracks wins, losses, and draws across three rounds to fairly determine the overall winner.

# Why This

01

## Learning:

AI adapts by playing against itself, improving with each game through self-play.

02

## Engagement:

Win, lose and draw create a fun and competitive experience.

03

## Simplicity:

A clean interface ensures intuitive and smooth gameplay for all users.

# Approach?



Thank

For Your Time And  
Attention Today.

You